

Lead (Pb) Contaminant Measurement on Distilled Water using Hybrid Sensors and Linear Regression

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Abstract— Heavy metals such as Pb, Hg, Cd, and As are commonly found in rivers and drinking water and can cause various health problems. This research aims to develop a linear regression model to predict the concentration of heavy metals (Pb) in distilled water based on TDS and pH sensors. Six experimental scenarios were utilized, each with different training data but the same testing data. A total of 30 rows of data were collected, each comprising five sets of data with Pb concentrations of 10 ppm, 20 ppm, 30 ppm, 40 ppm, 50 ppm, and 60 ppm. One data point from each concentration was randomly selected as test data. The rest are used to six form training data: raw, average, median, maximum, minimum, and skewness. The model's performance was evaluated using R^2 and RMSE. The analysis results indicated that the raw training data model achieved the best performance, with the highest R^2 value and the lowest RMSE. Further analysis showed that AVG_TDS had the most significant impact on predicting Pb concentrations.

Keywords—heavy metal, lead, linear regression, pH, sensor, TDS

I. INTRODUCTION

Water is a vital resource for human life and ecosystems. Water is important in various domestic, agricultural, and industrial activities. However, previous work showed that the five rivers in Indonesia had heavy metal pollution exceeding environmental quality standards [1]. Heavy metals such as Pb, Hg, Cd, and As are commonly found in river water [2] and drinking water [3].

The causes of lead (Pb) pollution vary greatly, ranging from industrial waste [4], agriculture, and residential areas to mining activities [5]. In addition, domestic waste from households, including domestic wastewater and waste from household products [6] containing Pb, can also pollute water

sources. More than 80% of the garbage produced by human activity is dumped into rivers and seas untreated, which pollutes the ecosystem [7]. Water pollution, mainly caused by Pb contaminants, has broad and varied impacts. From damage to aquatic ecosystems [8, 9], reduced clean water resources that humans can use, to increased risk of disease due to consumption of contaminated water [10]. Heavy metals such as Pb can accumulate in the human body and cause various health problems, including damage to the cardiovascular system, nervous system, kidneys, and even cancer [11]. Research conducted by Lin [7] also revealed that 80% of illnesses and 50% of child deaths worldwide are caused by poor water quality.

Serious efforts are required to tackle water pollution issues. One solution that can be implemented is utilizing sensor technology [12]. Sensors are devices capable of detecting and measuring specific environmental parameters [13]. Previous research uses Total Dissolved Solids (TDS) and pH sensors for water quality monitoring [14]. This research proposed Pb contaminant measurement on distilled water using TDS and pH Sensors.

TDS and pH sensors are used as initial indicators to assess the Pb contaminants of water samples. TDS sensors measure the total concentration of dissolved substances in water, including minerals, salts, and metals [15]. Meanwhile, pH sensors use the concentration of hydrogen ions to determine how acidic or alkaline water is [16]. Two output parameters were extracted from these sensors: TDS and pH values. Although the outputs of these two parameters do not directly indicate the concentration of Pb contamination, they serve as crucial indicators of water quality, as variations in the values of these parameters can indicate potential contamination or changes in the chemical composition of water.

This research used a linear regression method to analyze the relationship between TDS, pH, and Pb contamination concentrations. The regression model was also used to predict Pb contamination concentrations based on the measured TDS and pH values. The regression model's performance was evaluated using the coefficient of determination (R^2) and Root Mean Square Error (RMSE). This research aims to determine the concentrations of heavy metal lead (Pb) in distilled water by conducting regression analysis between the heavy metal concentrations and sensor outputs.

There are several sections in this paper: Section 2 presents the related works. Section 3 explains in detail the methods used in this research. Section 4 explains the research results. The final section contains the conclusions of this paper.

II. RELATED WORKS

Several previous studies evaluated water quality. One is research conducted by Luchian, et al. [17] in 2019. The study by Lucian et al. assessed the heavy metal content of five lakes in the Moldova area of Romania to determine the water quality. The effects of heavy metal pollution on ecosystems and human health are examined in that study, as well as the effects on water quality. One of the advantages of previous research lies in its comprehensive sampling methodology and rigorous analysis techniques. Luchian et al. collected water and sediment samples from five different lakes and performed Atomic Absorption spectrophotometry analysis to determine heavy metal concentrations. However, one of the limitations of previous research is that the researchers only used Atomic Absorption Spectrophotometry (AAS) as the main instrument for analysis. AAS requires testing in a laboratory, so it is impossible to conduct testing directly near the water sample location in real time. The research results in heavy metal concentrations were then compared with the maximum limits permitted by Romanian regulations. The research found that some lakes experienced pollution levels that exceeded the limits set for surface water, indicating a potential risk to human health and the surrounding environment.

Research related to water quality evaluation was also carried out by Pratama, et al. [14]. The main objective of previous research is to develop a prototype monitoring system that aims to assess the quality of drinking water by measuring two parameters, namely pH and Total Dissolved Solids (TDS). Using an integrated hardware design with the pH 201 C sensor and TDS SEN0244, the system successfully developed in previous research was able to provide good results. Tests show that the pH sensor has an accuracy rate of 96.63% with an average error of 3.367%, while the TDS sensor shows an accuracy rate of 99.10% with an average error of 0.90%. However, despite its accuracy, previous research does not detail the data processing process to determine whether drinking water is safe for consumption.

Lowongtrakool [18] also conducted similar research evaluating water quality. Lowongtrakool integrates sensors in an Internet of Things (IoT) approach to analyzing water quality. The research results show that the integrated sensor has a high accuracy and precision of up to 99%. The main advantage of previous research is its ability to collect water quality data accurately. However, other factors, such as the impact of room temperature and light intensity on the sensor, have not been considered.

TABLE I. ATTRIBUTE DESCRIPTION

Attribute	Description
Timestamp	The time when data was collected consisting of the date along with hours, minutes and seconds.
TDS_Value	Total Dissolved Solids (TDS) value in ppm (parts per million).
PH_Value	The pH value of the water.

The research carried out by Sudrajad, et al. [19] aims to develop a technological solution that can monitor and predict lake water quality in real time using the IoT approach and linear regression. By applying a linear regression approach, they achieved a high accuracy level with a Mean Absolute Percentage Error (MAPE) value of 3.57%. The main strength of this study is the integration of innovative technologies, including the use of LoRa networks and solar panels as energy sources, which makes the system efficient and sustainable.

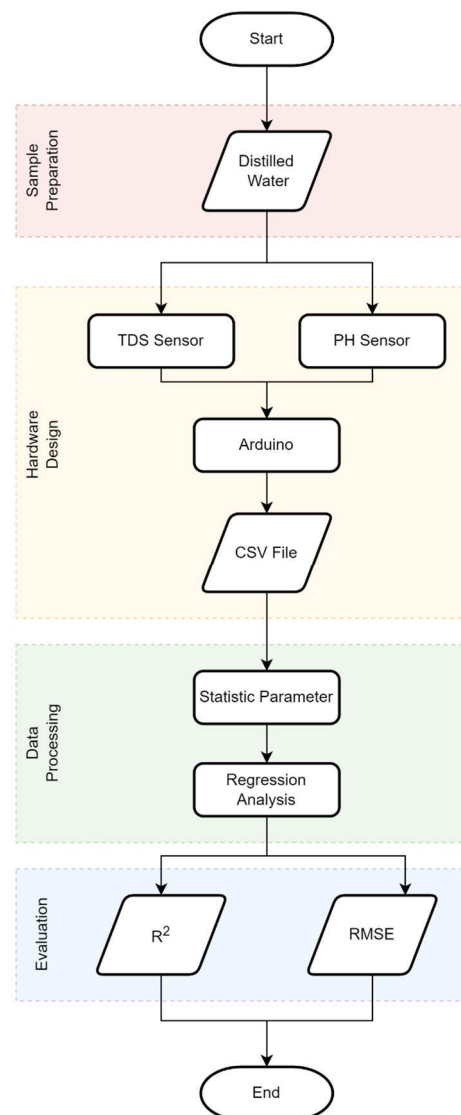


Fig. 1 Proposed Method

III. PROPOSED METHOD

This research consists of several stages: sample preparation, hardware design, data processing, and evaluation. Fig. 1 depicts the stages of this research.

A. Sample and Sensor Preparation

Water samples are prepared to be tested for Pb contaminant using a hybrid sensor system. This research uses several materials and solutions, such as distilled water, solid heavy metal Pb, and buffer solutions. Distilled water is the primary medium for all sample preparation, while buffer solutions are required for pH adjustment and sensor calibration.

Sample preparation begins with making a heavy metal stock solution of 100 ppm. The stock solution is then serially diluted to obtain six samples with concentrations of 10 ppm, 20 ppm, 30 ppm, 40 ppm, 50 ppm, and 60 ppm. Sensor calibration is a crucial process to ensure accuracy before measurements are taken. For calibration, distilled water without contaminants is used as the baseline. The TDS sensor is calibrated to show zero values in this baseline condition, while the pH sensor is adjusted to display a neutral value of seven. Calibration involves multiple iterations of adjusting the constant values in the microcontroller to align the sensor readings with known reference values. Additionally, the hybrid sensor system includes periodic checks using standard solutions to confirm that sensor drift is minimized over time. This rigorous calibration ensures reliable data acquisition during the experiments.

Data is collected by measuring sample properties using a TDS (Total Dissolved Solids) and pH sensors. TDS sensors measure the total concentration of dissolved solids in water, while pH sensors measure the acidity or alkalinity of a solution. These two sensor outputs will then be correlated with the presence of heavy metals. One test produces 1 CV file, which contains three columns: timestamp, TDS value (in ppm units), and PH value. Table I shows a description of each column or attribute.

B. Hardware Components

The hybrid sensor system consists of several main components: TDS sensor, pH sensor, Arduino Uno microcontroller, and computer (PC) for data processing. The following is an explanation and picture of each principal component:

1) TDS sensors

TDS sensors are used to measure total dissolved solids in water, which can indicate the presence and concentration of heavy metals. This sensor measures the electrical conductivity of a solution to determine the amount of dissolved solids.

2) pH sensor

pH sensors are used to measure the acidity or alkalinity of water, which affects the solubility and behavior of heavy metals. This sensor is vital to ensure that the pH of the water is within a range that can affect the proper detection of heavy metals.

3) Arduino Uno microcontroller

An Arduino Uno microcontroller collects and processes data from TDS and pH sensors. The Arduino acts as an

interface between the sensor and the computer, sending data read from the sensor to the computer for further analysis.

4) Computer (PC)

A computer is used to process data collected by Arduino from TDS and pH sensors.

C. Data Analysis

The average (mean) and standard deviation are calculated for each parameter in 1 set of measurements to analyze sensor data. The average is calculated using the formula described in Eq. (1), and the standard deviation is calculated using the formula explained in Eq. (2) [20].

$$\bar{x} = \frac{\sum x}{n} \quad (1)$$

$$SD = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} \quad (2)$$

where x is the value of each observational data, $\bar{x}$ is the mean value of x , n is the amount of data, and SD is the standard deviation.

In applying the regression concept, this research utilizes linear regression to analyze the relationship between Pb concentrations in water samples and hybrid sensor readings (TDS and pH). Linear regression determines the relationship between sensor output and contaminant concentration, with the formula shown in Eq. (3).

$$Y = a + bX \quad (3)$$

where y is the dependent variable, x is the dependent variable, a is the intercept or constant, and b is the regression coefficient or slope [21].

D. Evaluation

The performance of the regression model is then evaluated using the evaluation metrics coefficient of determination (R^2) [21] and Root Mean Square Error (RMSE) [22]. R^2 is a coefficient of determination that measures the proportion of variability in the dependent variable that the independent variables in the model can explain. RMSE is a metric that measures the average error between the value predicted by the model and the actual value. Eq. (4) shows the R^2 formula, while Eq. (5) shows the RMSE formula.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the average value of the actual value, and n is the number of observations [23]. The R^2 value ranges between 0 and 1. A R^2 value close to 1 indicates that the model can explain most of the variation in the data, while a value close to 0 indicates that the model explains slight variation.

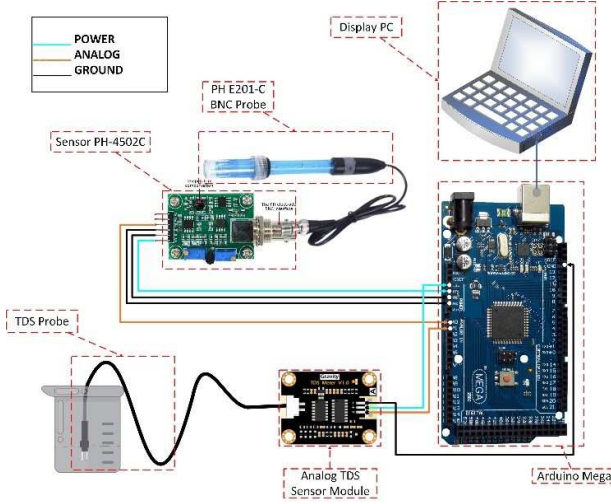


Fig. 2 Pb Measurement System using Hybrid Sensor

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and n is the number of observations [23]. A lower RMSE number denotes a lesser model error and a higher measurement accuracy. The model's error increases with the increase in the RMSE value.

IV. RESULT AND DISCUSSION

A. Hardware System

The hardware system includes a TDS sensor and a pH sensor. Data is collected by inserting a TDS sensor probe and a PH sensor probe into the sample. The TDS and pH sensors are waterproof and can continuously monitor the aquatic environment. TDS and pH sensors produce output as an analog voltage, which is then converted to digital voltage using an analog-to-digital converter (ADC) before the microcontroller can process the data. The microcontroller used in this research is Arduino Mega. The TDS sensor has an output range of 0-2.3V, while the pH sensor has an output range of 0-5V. These sensors require a 3.3V and 5V power supply, according to the power available on the Arduino Mega board. The Arduino Mega plugs into a PC's USB port to collect data and stream it in Excel Data Streamer, aiding in data exploration and data preparation. Fig. 2 shows the integration between components in the heavy metal measurement system in water.

B. Data Collection and Description

The sample testing process was conducted 30 times. The tests were divided as follows: five times using water samples containing 10 ppm Pb, five times using samples with 20 ppm, five times with 30 ppm, five times with 40 ppm, five times with 50 ppm, and five times with 60 ppm. Each test produced a CSV file containing three columns, as Table I shows. The

TABLE II. TOP 10 DATA

ID	AVG_TDS	AVG_PH	STD_TDS	STD_PH	Pb Concentration
1	5,4180	5,7042	0,3442	0,1483	10
2	5,5100	5,7280	0,0000	0,0823	10
3	5,5100	5,7610	0,0000	0,1342	10
4	4,4980	5,7647	0,6102	0,0691	10
5	4,1300	5,7377	0,0000	0,0871	10
6	6,7793	6,5558	0,3392	0,1289	20
7	6,5980	6,5648	0,5440	0,2049	20
8	6,8700	6,4790	0,0000	0,1014	20
9	6,8700	6,5017	0,0000	0,1141	20
10	6,8700	6,2448	0,0000	0,1847	20

average and standard deviation of the test results are calculated for each column. Each test's averages and standard deviations were combined into a CSV file. This combined CSV file, consisting of 6 attribute columns and 30 rows of data, will serve as the primary input in the regression analysis process. Table II shows the top 10 data entries from the combined CSV file of the test results.

C. Result of Linear Regression and Evaluation

A regression model was constructed for this study in order to investigate the correlation between independent variables and dependent variables. The independent variables referred to in this research are four attributes, namely AVG_TDS, AVG_PH, STD_TDS, and STD_PH. Meanwhile, the dependent variable is Concentration_Heavy_Metal, which is the prediction target in this analysis.

This research uses six regression experiment scenarios, each with different training data but the same testing data, to determine best-fit training data. Descriptive statistics, such as average, median, maximum, minimum, and skewness, extract information from training data and become experiment scenarios. There are 30 rows of data, consisting of five data sets for each Pb concentration: 10 ppm, 20 ppm, 30 ppm, 40 ppm, 50 ppm, and 60 ppm. One data from each Pb concentration is taken randomly, producing six rows of test data. The remaining 24 data are used to form six types of training data as follows:

- (1) Training Data 1: Contains 24 raw data without modification.
- (2) Training Data 2: Contains 6 average data results from 4 data for each Pb concentration.
- (3) Training Data 3: Contains 6 median data results from 4 data for each Pb concentration.
- (4) Training Data 4: Contains 6 maximum data from 4 data for each Pb concentration.
- (5) Training Data 5: Contains 6 minimum data from 4 data for each Pb concentration.
- (6) Training Data 6: Contains 6 skewness data from 4 data for each Pb concentration.

TABLE III. EVALUATION RESULTS

Scenario_ID	Training Data	R ²	RMSE
1	Training Data 1	0.976	2.683
2	Training Data 2	0.822	7.207
3	Training Data 3	0.869	6.176
4	Training Data 4	0.857	6.456
5	Training Data 5	0.757	8.422
6	Training Data 6	-1088.725	563.770

After forming training and testing data, linear regression analysis is performed for each scenario. Model evaluation is carried out using two metrics: R² and RMSE. The evaluation results of the six scenarios are presented in Table III.

The evaluation results show that the linear regression model using Training Data 1 performs better than the other five scenarios because the raw data better represents the variations in the test data. Almost all (75%) of the values in the test data are within the range of the training data, which indicates that the model built with Training Data 1 is more capable of predicting Pb concentrations accurately.

In contrast, the scenario using Training Data 6 exhibits significantly poor performance, as indicated by the highly negative R² value of -1088.725 and an exceptionally high RMSE of 563.770. The negative R² value suggests that the model fails to predict the data accurately and is worse than a horizontal line (which would have an R² of 0). That typically occurs when there is a severe mismatch between the training and test data, leading to the model making poor predictions.

The high RMSE further confirms that the model's predictions are far from the actual values, indicating a significant degree of error. This discrepancy can arise due to several factors. First, the training data may not effectively represent the basic patterns of the test data, potentially due to inadequate sample diversity or uneven distribution of concentrations in the dataset. Second, sensor inaccuracies during data acquisition could contribute to noisy input data, particularly if sensor drift or incomplete calibration occurred during the experimental process. Third, environmental factors such as temperature fluctuations or interference from other ions in the solution could affect sensor performance, introducing variability into the data.

Other scenarios, such as those using Training Data 2, 3, 4, and 5, show moderate performance with R² values ranging from 0.757 to 0.869 and RMSE values between 6.176 and 8.422. These results indicate that while these models have some predictive capability, they are not as accurate as those using Training Data 1. The variation in performance among these scenarios could be attributed to differences in the representativeness and quality of the training data used. The choice of training data significantly impacts the performance of the regression model. Training Data 1 provides the best fit for the test data. At the same time, Training Data 6 demonstrates the worst performance, highlighting the importance of selecting appropriate and representative training data for building accurate predictive models.

OLS Regression Results						
Dep. Variable:	Lead (Pb)	R-squared:	0.970			
Model:	OLS	Adj. R-squared:	0.964			
Method:	Least Squares	F-statistic:	153.6			
Date:	Thu, 15 Aug 2024	Prob (F-statistic):	3.48e-14			
Time:	15:31:15	Log-Likelihood:	-60.082			
No. Observations:	24	AIC:	130.2			
Df Residuals:	19	BIC:	136.1			
Df Model:	4					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	35.0000	0.679	51.578	0.000	33.580	36.420
x1	15.8120	1.138	13.901	0.000	13.431	18.193
x2	-1.0907	1.139	-0.957	0.350	-3.476	1.294
x3	0.5679	0.689	0.825	0.420	-0.873	2.009
x4	1.0311	0.687	1.501	0.150	-0.406	2.468
Omnibus:	0.227	Durbin-Watson:	0.590			
Prob(Omnibus):	0.893	Jarque-Bera (JB):	0.413			
Skew:	-0.149	Prob(JB):	0.813			
Kurtosis:	2.431	Cond. No.	3.04			

Fig. 3 Linear Regression Result using Training Data 1

As the best result, Training Data 1 is used for further analysis. As shown in Fig 3, the R² score model achieves 0.97, which means independent variables can explain 97% variation in a dependent variable. The residual test using Durbin-Watson showed positive autocorrelation, which can cause problems in prediction. The Jarque-Bera test shows the residual already normal distribution, which supports the normality assumption in linear regression. A significant p-value explains the relationship between Pb contamination, TDS value, and pH value. Only one variable, AVG_TDS, significantly influences pB concentration in distilled water. The STD_TDS, AVG_PH, and STD_PH variables are not statistically significant in the measurement pB contaminant model. Based on the conducted analysis, the linear regression equation for predicting pB contaminant in distilled water based on the TDS sensor and pH sensor is represented as Eq. (6).

$$y = 35.0000 + 15.8120 \times \text{AVG_TDS} - 1.0907 \times \text{AVG_PH} + 0.5679 \times \text{STD_TDS} + 1.0311 \times \text{STD_PH} \quad (6)$$

V. CONCLUSION

This research successfully developed a linear regression model to predict the concentration of heavy metals (Pb) in distilled water samples based on TDS and pH sensor outputs. Among the six experimental scenarios, the model using raw training data (24 unmodified data points) showed the best performance with the highest R² value and the lowest RMSE. Unmodified training data (raw data) are more accurate than average, median, maximum, minimum, or skewness-modified data. Moreover, the significance of each variable was analyzed, and AVG_TDS was found to have the most substantial impact on pB concentration predictions. Conversely, pH variables are not statistically significant in the measurement pB contaminant model.

The developed linear regression model shows significant potential for practical applications in real-time water quality monitoring systems. By integrating this model into IoT-based platforms, such systems can provide continuous and automated detection of Pb contamination in water sources, offering timely warnings and decision-making support for environmental monitoring agencies and industrial

stakeholders. However, the model's identified positive autocorrelation may cause prediction instability in certain scenarios. The autoregressive (AR) model can employ future work to eliminate the problem.

ACKNOWLEDGMENT

This research was funded by Institut Teknologi Sepuluh Nopember (ITS) under Penelitian Dana Departemen Program, Penelitian Keilmuan, Penelitian Flagship, Penelitian Kolaborasi Pusat and under the Project Scheme of the Publication Writing and Intellectual Property Rights (IPR) Incentive (Penulisan Publikasi dan Hak Kekayaan Intelektual/PPHKI) Program 2024.

REFERENCES

- [1] A. Yusfaddillah, R. Dwi Saputri, T. W. Edelwis, and H. Pardi, "Heavy Metal Pollution in Indonesian Waters," *BIO Web Conf*, vol. 79, p. 04001, Dec. 2023, doi: 10.1051/bioconf/20237904001.
- [2] Q. Zhou, N. Yang, Y. Li, B. Ren, X. Ding, H. Bian, and X. Yao, "Total concentrations and sources of heavy metal pollution in global river and lake water bodies from 1972 to 2017," *Glob Ecol Conserv*, vol. 22, p. e00925, Jun. 2020, doi: 10.1016/j.gecco.2020.e00925.
- [3] W. Dong, Y. Zhang, and X. Quan, "Health risk assessment of heavy metals and pesticides: A case study in the main drinking water source in Dalian, China," *Chemosphere*, vol. 242, p. 125113, Mar. 2020, doi: 10.1016/j.chemosphere.2019.125113.
- [4] M. Zhang, X. Sun, and J. Xu, "Heavy metal pollution in the East China Sea: A review," *Mar Pollut Bull*, vol. 159, p. 111473, 2020, doi: <https://doi.org/10.1016/j.marpolbul.2020.111473>.
- [5] F. T. Gameda, D. D. Guta, F. S. Wakjira, and G. Gebresenbet, "Occurrence of heavy metal in water, soil, and plants in fields irrigated with industrial wastewater in Sabata town, Ethiopia," *Environmental Science and Pollution Research*, vol. 28, no. 10, pp. 12382–12396, Mar. 2021, doi: 10.1007/s11356-020-10621-6.
- [6] H. S. Budi, M. J. Catalan Oplencia, A. Afra, W. K. Abdelbasset, D. Abdullaev, A. Majdi, M. Taherian, H. A. Ekrami, and M. J. Mohammadi, "Source, toxicity and carcinogenic health risk assessment of heavy metals," *Rev Environ Health*, vol. 39, no. 1, pp. 77–90, Mar. 2024, doi: 10.1515/reveh-2022-0096.
- [7] L. Lin, H. Yang, and X. Xu, "Effects of Water Pollution on Human Health and Disease Heterogeneity: A Review," *Front Environ Sci*, vol. 10, Jun. 2022, doi: 10.3389/fenvs.2022.880246.
- [8] P. Jin, J. Zhang, J. Wan, S. Overmans, G. Gao, M. Ye, X. Dai, J. Zhao, M. Xiao, and J. Xia, "The Combined Effects of Ocean Acidification and Heavy Metals on Marine Organisms: A Meta-Analysis," *Front Mar Sci*, vol. 8, Dec. 2021, doi: 10.3389/fmars.2021.801889.
- [9] D.-P. Häder, A. T. Banaszak, V. E. Villafañe, M. A. Narvarte, R. A. González, and E. W. Helbling, "Anthropogenic pollution of aquatic ecosystems: Emerging problems with global implications," *Science of The Total Environment*, vol. 713, p. 136586, 2020, doi: <https://doi.org/10.1016/j.scitotenv.2020.136586>.
- [10] U. Okerefor, M. Makhatha, L. Mekuto, N. Uche-Okerefor, T. Sebola, and V. Mavumengwana, "Toxic Metal Implications on Agricultural Soils, Plants, Animals, Aquatic life and Human Health," *Int J Environ Res Public Health*, vol. 17, no. 7, p. 2204, Mar. 2020, doi: 10.3390/ijerph17072204.
- [11] V. Kumar, R. D. Parihar, A. Sharma, P. Bakshi, G. P. Singh Sidhu, A. S. Bali, I. Karaouzas, R. Bhardwaj, A. K. Thukral, Y. Gyasi-Agyei, and J. Rodrigo-Comino, "Global evaluation of heavy metal content in surface water bodies: A meta-analysis using heavy metal pollution indices and multivariate statistical analyses," *Chemosphere*, vol. 236, p. 124364, Dec. 2019, doi: 10.1016/j.chemosphere.2019.124364.
- [12] A. Bria, G. Cerro, M. Ferdinandi, C. Marrocco, and M. Molinara, "An IoT-ready solution for automated recognition of water contaminants," *Pattern Recognit Lett*, vol. 135, pp. 188–195, Jul. 2020, doi: 10.1016/j.patrec.2020.04.019.
- [13] S. L. Ullo and G. R. Sinha, "Advances in Smart Environment Monitoring Systems Using IoT and Sensors," *Sensors*, vol. 20, no. 11, p. 3113, May 2020, doi: 10.3390/s20113113.
- [14] I. P. E. W. Pratama, S. F. Mujiyanti, P. Y. Aisyah, E. A. Siahaan, and M. T. Algifary, "A Novel PH and TDS Monitoring System in Drinking Water," In: *2023 International Conference on Advanced Mechatronics, Intelligent Manufacture and Industrial Automation (ICAMIMIA)*, Nov. 2023, pp. 1–5. doi: 10.1109/ICAMIMIA60881.2023.10427721.
- [15] G. E. Adjovu, H. Stephen, D. James, and S. Ahmad, "Measurement of Total Dissolved Solids and Total Suspended Solids in Water Systems: A Review of the Issues, Conventional, and Remote Sensing Techniques," *Remote Sens (Basel)*, vol. 15, no. 14, p. 3534, Jul. 2023, doi: 10.3390/rs15143534.
- [16] L. Manjakkal, D. Szwagierczak, and R. Dahiya, "Metal oxides based electrochemical pH sensors: Current progress and future perspectives," *Prog Mater Sci*, vol. 109, p. 100635, Apr. 2020, doi: 10.1016/j.pmatsci.2019.100635.
- [17] C. E. Luchian, A. Popirda, L. C. Colibaba, A. Georgiana Popa, C. E. Scuturasu, L. Rotaru, and V. V. Cotea, "Quantitative Determination of Heavy Metal in Water and Sediment from Lakes in North Moldova, Romania," In: *2019 E-Health and Bioengineering Conference (EHB)*, Nov. 2019, pp. 1–4. doi: 10.1109/EHB47216.2019.8969910.
- [18] C. Lowongtrakool and P. Lorwongtrakool, "IoT Based Water Quality Measurement Using Hybrid Sensors and Data Mining," In: *2018 International Conference on Information Technology (InCIT)*, Oct. 2018, pp. 1–6. doi: 10.23919/INCIT.2018.8584873.
- [19] A. Sudrajad, H. H. Nuha, and S. Novian Anggis, "Monitoring and Prediction of Water Levels in Telkom University Techno Lake Based on the Internet of Things using the Linear Regression Approach," In: *2023 International Conference on Artificial Intelligence, Blockchain, Cloud Computing, and Data Analytics (ICoABCD)*, Nov. 2023, pp. 19–24. doi: 10.1109/ICoABCD59879.2023.10390947.
- [20] A. Hussain and A. Aslam, "Cardiovascular Disease Prediction Using Risk Factors: A Comparative Performance Analysis of Machine Learning Models," *Journal on Artificial Intelligence*, vol. 6, no. 1, pp. 129–152, 2024, doi: 10.32604/jai.2024.050277.
- [21] S. Rath, A. Tripathy, and A. R. Tripathy, "Prediction of new active cases of coronavirus disease (COVID-19) pandemic using multiple linear regression model," *Diabetes & Metabolic Syndrome: Clinical Research & Reviews*, vol. 14, no. 5, pp. 1467–1474, Sep. 2020, doi: 10.1016/j.dsx.2020.07.045.
- [22] B. Mohammadi and S. Mehdizadeh, "Modeling daily reference evapotranspiration via a novel approach based on support vector regression coupled with whale optimization algorithm," *Agric Water Manag*, vol. 237, p. 106145, Jul. 2020, doi: 10.1016/j.agwat.2020.106145.
- [23] D. Rahmawati and R. Sarno, "Anomaly detection using control flow pattern and fuzzy regression in port container handling," *Journal of King Saud University - Computer and Information Sciences*, vol. 33, no. 1, pp. 11–20, Jan. 2021, doi: 10.1016/j.jksuci.2018.12.004.